

TU059/TU060/TU256/PhD Probability and Statistical Inferences
PSI CA Part Three Marking Scheme/Rubric

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Marking Breakdown

ASPECT	ISSUES CONSIDERED				OVERALL MARKS
HYPOTHESES	- Correctly develop and present a relevant hypothesis for tasks 1, 2 and 3 (3 x 3 marks) - Note: At least 3 hypotheses required (linear regression, logistic regression, comparison) ¹				9 marks
DIMENSION REDUCTION	- Describes the conduct of a dimension reduction technique describing the concepts of interest generating all appropriate statistics and illustrations. - Critically discussion of effectiveness of the dimension reduction technique including description of the outcomes.				8 marks 10 marks
LINEAR REGRESSION	- Briefly present the concepts used in the model. - Correctly conduct the building of a model and the assessment the fit and usefulness generating all appropriate statistics and illustrations.				5 marks 10 marks
	Critically discuss the model outcomes, assessment of fit and usefulness, illustration of usefulness referencing all appropriate statistics and illustrations.				15 marks
LOGISTIC REGRESSION	- Briefly present the concepts used in the model. - Correctly conduct the building of a model and the assessment the fit and usefulness generating all appropriate statistics and illustrations.				5 marks 10 marks
	Critically discuss the model outcomes, assessment of fit and usefulness, illustration of usefulness referencing all appropriate statistics and illustrations.				15 marks
COMPARISON MODEL	- Briefly present the predictor added/removed. - Correctly conduct the building of a model and the assessment the fit and usefulness generating all appropriate statistics and illustrations.				2 marks 5 marks
	Critically discuss the model outcomes, assessment of fit and usefulness, illustration of usefulness referencing all appropriate statistics and illustrations.				6 marks
	Total Marks Available ²				100 marks
ACHIEVEMENT	EXCELLENT	GOOD	SATISFACTORY	UNSATISFACTORY	
% OF MARKS AVAILABLE	> 75%	60-80%	40-60%	<40%	

¹ At least one of your hypotheses for regression must involve a differential effect and therefore at least one of your models needs to include a nominal variable as a predictor.

² The task will be marked out of 100 and your result will be weighted to reflect the 60% of this task in the calculation of the overall module marks.

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Key Characteristics associated with each achievement level for each aspect are described on the next pages.³

HYPOTHESES Rubric

HYPOTHESES			
KEY CHARACTERISTICS OF ACHIEVEMENT LEVELS			
EXCELLENT	SATISFACTORY	BASIC	UNSATISFACTORY
• Correctly states hypotheses in a manner that supports investigation using the relevant technique. • 3 hypotheses included.	• States hypotheses for each technique with some minor errors, but which are still suitable for investigation using the relevant technique. • 3 hypotheses included.	• States hypotheses for each technique with some minor errors or omissions, but which are still suitable for investigation using the relevant technique. • 3 hypotheses included.	• Hypotheses are either not included or contain significant errors which render them useless.

³ The characteristics described below in the rubric are for a student submission which addresses all components of the assignment. Where a student attempts fewer components, the marks achieved will depend on how many questions have been addressed and the level of achievement for each question. For example, if a student attempts only THREE components, but completes everything to the level indicated by the category of EXCELLENT, their mark will not reach the level associated with this category but with a lower a category.

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DIMENSION REDUCTION Rubric

DIMENSION REDUCTION KEY CHARACTERISTICS OF ACHIEVEMENT LEVELS			
EXCELLENT	SATISFACTORY	BASIC	UNSATISFACTORY
• Correctly reports on the examination of the variables chosen to determine reliability. • Correctly describes the conduct with relevant statistics and presenting the relevant outcomes. • Critically discusses the reliability, implications and choices resulting. • Critically examines the results of the checks to establish the data is suitable, implications and choices resulting. • Critically examines the findings and how these influenced the choice of factors to retain. • Describes the factors/components extracted and explains choices made. • Derives correct conclusions for the related hypothesis and describes factors/components extracted.	• Correctly examines the variables chosen to determine reliability. • Correctly presents the outcomes of checks to establish the data is suitable generating appropriate statistics. • Discusses the reliability, implications and choices resulting with some minor omissions/errors. • Discusses the results of the checks to establish the data is suitable, examining some implications and choices resulting with some minor omissions/errors. • Correctly describes the conduct with relevant statistics and presenting the relevant outcomes with minor errors in interpretation. • Describes the factors/components extracted with an effort to explain choices made with some minor errors/omissions which impact the conclusions drawn.	• Correctly presents the outcome of checks to establish the data is suitable generating appropriate statistics with some errors/omissions. • Correctly describes the conduct with relevant statistics and presenting the relevant outcomes with some minor errors/omissions. • Presents the reliability analysis for the variables chosen and discusses the implications with some errors. • Describes the factors/components extracted.	• Either no reliability analysis or significant errors in the reporting of the reliability analysis. • Presents some of the checks to establish the data is suitable generating appropriate statistics/presents the checks with errors. • Presents some relevant statistics and graphs from the conduct of the dimension reduction – errors obvious in the use of the outcomes. • States some factors/components extracted – errors obvious in the choice.

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LINEAR REGRESSION Rubric

Linear Regression KEY CHARACTERISTICS OF ACHIEVEMENT LEVELS			
EXCELLENT	SATISFACTORY	BASIC	UNSATISFACTORY
• Correctly derives evidence for including predictors. • Correctly creates the model. • Conducts the relevant checks for assumptions. • Evidence for predictors is illustrated using appropriate statistics and visuals. • Illustrates the findings for relevant sample data. • Correctly presents relevant statistics on the fit and usefulness of the model overall and the co-efficients. • Presents the findings of assumptions checks using appropriate statistics and visuals. • Critically discusses implications of assessment of assumptions. • Critically discusses the implications of the findings (from the model and assumptions) for the related hypothesis when drawing a conclusion.	• Correctly derives evidence for including predictors. • Correctly creates the model. • Conducts the relevant checks for assumptions. • Evidence for predictors is illustrated using appropriate statistics and visuals. • Presents relevant statistics on the fit and usefulness of the model overall and the co-efficients with some minor omissions/errors. • Illustrates the findings for relevant sample data with some minor omissions/errors. • Presents the findings using appropriate statistics and visuals. • An effort is made to discuss the implications but omissions/errors impact level of criticality. • States a conclusion for related hypothesis.	• Derives evidence for including predictors with some minor omissions. • Correctly creates the model. • A basic effort to examine some of the assumptions supported by relevant statistics. • Evidence for predictors is illustrated using appropriate statistics and visuals with minor omissions. • Presents relevant statistics on the fit and usefulness of the model overall and the co-efficients with some minor omissions. • A basic effort to illustrate the findings. • Presents some elements of statistics to support assumptions checks. • States a conclusion for related hypothesis.	• Does not derive sufficient evidence for including predictors. • Creates the model but does not examine or present the fit and usefulness of the model appropriately. • Evidence for including predictors is illustrated using appropriate statistics and visuals with errors/omissions. • Does not examine assumptions or has major omissions/errors. • No effort to illustrate the usefulness of the model. • No effort to draw conclusion for the related hypothesis.

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LOGISTIC REGRESSION Rubric

LOGISTIC REGRESSION KEY CHARACTERISTICS OF ACHIEVEMENT LEVELS			
EXCELLENT	SATISFACTORY	BASIC	UNSATISFACTORY
- Clearly presents evidence for including predictors illustrated using appropriate statistics and visuals. - Correctly creates the model and derives relevant statistics. - Clearly presents evidence for including predictors illustrated using appropriate statistics and visuals. - Presents relevant statistics on the fit and usefulness of the model overall and the co-efficients. - Conducts the relevant checks for assumptions and presents the findings using appropriate statistics and visuals. - Illustrates the findings for relevant sample data. - Critically discusses implications of assessment of assumptions. - Critically discusses the implications of the findings (from the model and assumptions) for the related hypothesis when drawing a conclusion.	- Clearly presents evidence for including predictors illustrated using appropriate statistics and visuals. - Correctly creates the model and presents relevant statistics on the fit and usefulness of the model overall and the co-efficients with some minor omissions/errors. - Conducts the relevant checks for assumptions and presents the findings using appropriate statistics and visuals with some minor omissions/errors. - Illustrates the findings for relevant sample data with some minor omissions/errors. - An effort is made to discuss the implications but omissions/errors impact level of criticality. - States a conclusion for related hypothesis.	- Presents evidence for including predictors illustrated using appropriate statistics and visuals with some minor omissions. - Correctly creates the model and presents relevant statistics on the fit and usefulness of the model overall and the co-efficients with some minor omissions. - A basic effort to illustrate the findings. - A basic effort to examine some of the assumptions supported by relevant statistics. - States a conclusion for related hypothesis.	- Does not present sufficient evidence for including predictors illustrated using appropriate statistics and visuals. - Creates the model but does not examine or present the fit and usefulness of the model appropriately. - No effort to illustrate the usefulness of the model. - No effort to draw conclusion for the related hypothesis.

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COMPARISON MODEL Rubric

COMPARISON MODEL KEY CHARACTERISTICS OF ACHIEVEMENT LEVELS			
EXCELLENT	SATISFACTORY	BASIC	UNSATISFACTORY
- Addresses all the criteria for the original model as outlined above for the EXCELLENT level. - PLUS - Discusses the inclusion/removal of predictor(s) presenting appropriate statistics and visuals if including. - Critically compares the fit and usefulness and assumptions checks of the model to the baseline model (created in task 2/3 as appropriate). - Critically discusses the implications of the findings (from the comparison) for the related hypothesis when drawing a conclusion.	- Addresses all the criteria for the original model as outlined above to the SATISFACTORY level. - PLUS - Discusses the inclusion/removal of predictor(s) presenting appropriate statistics and visuals if including. - Compares the fit and usefulness of the model to the baseline model (created in task 2/3 as appropriate) with some minor errors/omissions which limits criticality. - States a conclusion for related hypothesis.	- Addresses all the criteria for the original model as outlined above to the BASIC level. - PLUS - States the predictor(s) included/removed. - States differences with baseline model (created in task 2/3 as appropriate) but no discussion included. - States a conclusion for related hypothesis.	- Addresses all the criteria for the original model as outlined above to the UNSATISFACTORY level. - PLUS - No effort to compare model with the baseline model (created in task 2/3 as appropriate). - No effort to draw conclusion for the related hypothesis.